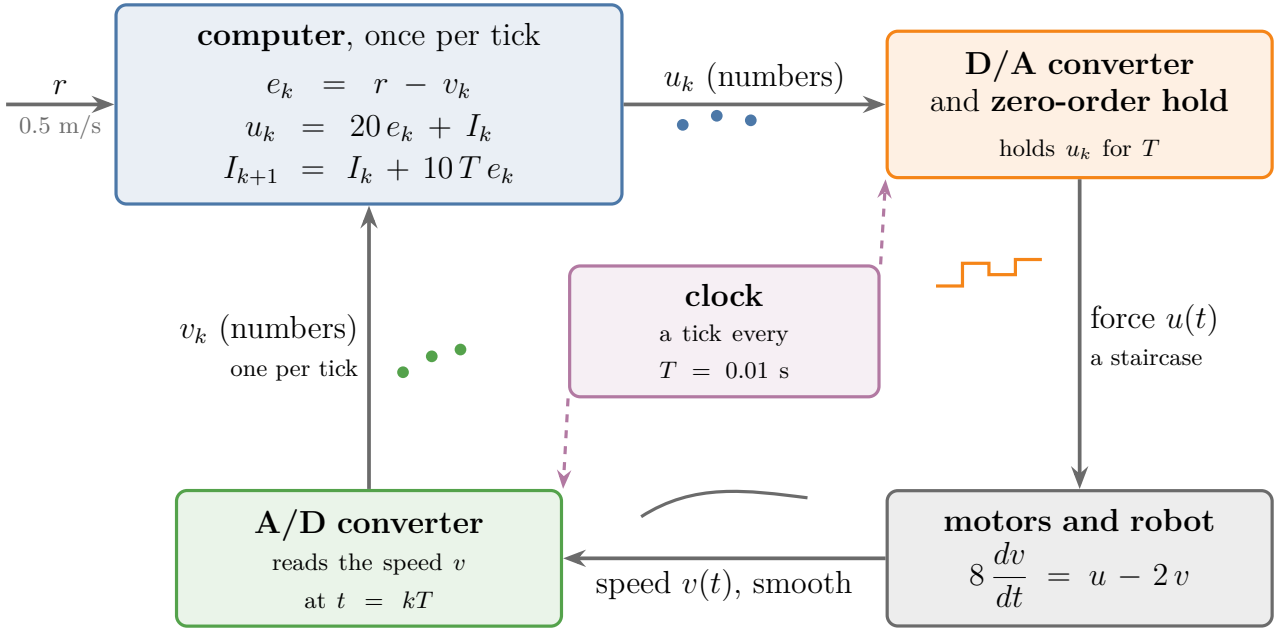

computer, once per tick
$e_k = r - v_k$
$u_k = 20\,e_k + I_k$
$I_{k+1} = I_k + 10\,T\,e_k$
r
0.5 m/s
u_k (numbers)
D/A converter
and zero-order hold
holds u_k for T
force $u(t)$
a staircase
motors and robot
$8\,\frac{dv}{dt} = u - 2\,v$
speed $v(t)$, smooth
A/D converter
reads the speed v
at $t = kT$
v_k (numbers)
one per tick
clock
a tick every
$T = 0.01$ s